

ECE 447

Lesson 33

GR2 Review Day



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

- **Schedule.**
 - Lesson 33 - **Review Day**
 - Lesson 34 - GR2 (Thursday, 13 Nov)
 - Lessons 35-37 - Binary Digital System Performance, Error Correction
 - Lesson 38 - MATLAB Lab 7: Matched filters, multi-path, OFDM, BER (substitute for Lt Col TBD)
 - Lesson 39 - Advanced topics: OFDM, MIMO, CDMA
 - Lesson 40 - Course review
- **Admin**
 - **HW7.** Due 10 Nov (Lsn 33) to Gradescope.

CHP 5 REVIEW: LESSONS 18-20, HWs 3 & 4

• Sampling

- Impulse train in time $\leftrightarrow$ impulse train in frequency
- Creates copies of message spectrum in frequency
- Sinc is ideal interpolator, if $f_s \geq 2B$
- Non-ideal pulses have "filtering" effect on sampled signal - equalizers reverse the filtering to obtain original signal
- What two requirements do we have for *distortionless* filtering?
- $G(f)e^{-j2\pi ft_0} = E(f)\tilde{G}(f) = E(f)P(f)\frac{1}{T_s}\sum_n G(f - nf_s)$, so make equalizer transfer function

$$E(f) = \begin{cases} T_s e^{-j2\pi ft_0} \cdot |P(f)|^{-1}, & |f| \leq B \\ \text{Flexible}, & B < |f| < f_s - B \\ 0, & |f| \geq f_s - B \end{cases}$$

- Just need to invert $P(f)$ and plug into $E(f)$! Example: 5.1-5a - find $E(f)$ if $p(t) = \Pi(\frac{t}{T_s} - \frac{1}{2})$

CHP 5 REVIEW: LESSONS 18-20, HWs 3 & 4

- PCM

- Just sample, quantize, and encode
- Message peak at $\pm m_p$, so each bin is $2m_p/L$, where $L = 2^n$ levels, n = number of bits
- Each additional bit improves the SNR how much?
- Max data rate per bandwidth is $\frac{2\text{bps}}{\text{Hz}}$ - assuming a noiseless channel w/ bandwidth B
- **If sampling exactly at Nyquist**, $f_s = 2B$, then min req'd channel bandwidth is $B_T = nB$
- **If sampling at greater than Nyquist**, $f_s > 2B$, then min req'd channel bandwidth is $B_T = \frac{nf_s}{2}$ or simply the data rate nf_s divided by 2
- Be able to solve problems like 5.2-3 or Example 5.2, where a max quantization error is given as a percentage of m_p and f_s is given as a percent of the Nyquist rate

CHP 6 REVIEW: LESSONS 22-28, HWs 5 & 6

- **Digital Communications**

- **Line codes**

- Be able to plot $y(t) = \sum_k a_k p(t - kT_b)$, i.e., the baseband waveform
 - Be able to use generic PSD equation in (6.11c)? Maybe, but I'd provide values for R_n
 - Band-limited vs. time-limited
 - **ISI**
 - Nyquist's First Criterion for Zero ISI:

$$p(t) = \begin{cases} 1, & t = 0 \\ 0, & t = \pm nT_b \end{cases}$$

- Excess bandwidth f_x , roll-off factor r
 - GR will **clearly** use R_b for bit rate and R_s for symbol rate/ baud
 - How do we accomplish timing synchronization?
 - Parts of eye diagrams

CHP 7 REVIEW: LESSONS 29-30, HW 7

- **Probability**

- Bernoulli trials using bits
- Conditional/total probability
- Gaussian RV, Q function
- Expected value, mean square, and variance
- Central limit theorem